

MECHANISM OF MEMORY

Headings

- Definition / Basic concept
- Molecular mechanisms
- Long-term potentiation (LTP)
- Neural mechanisms / Brain areas involved
- Habituation
- Sensitization
- Classical conditioning
- Post-tetanic potentiation
- Declarative and procedural memory
- Role of cholinergic neurons

1. Definition / Basic Concept

- Learning and memory are established by **long-lasting changes in synaptic function and neuronal connections**.
- Important neurochemical changes include:
 - ↑ Neurotransmitter release
 - Formation of **intracellular second messengers**
 - **Protein phosphorylation**
 - **Protein synthesis**
 - **Gene activation**
 - Formation/strengthening of **synaptic connections**

2. Molecular Mechanisms

Repeated stimulation

↓

Intracellular second-messenger formation

↓

Protein phosphorylation

↓

Altered neuronal excitability



Protein synthesis & gene activation



Strengthening / formation of synaptic connections



Learning & memory

3. Long-Term Potentiation (LTP) ★

- **LTP** = long-lasting increase in **synaptic strength** following repeated/high-frequency stimulation.
- It is an important mechanism for **learning and memory**.
- Mainly studied in the **hippocampus**.

Repeated / high-frequency stimulation



↑ **Intracellular Ca^{2+}**



Protein phosphorylation



↑ **Neuronal excitability & synaptic strength**



Protein synthesis



Formation/strengthening of synaptic connections



Long-term memory

4. Neural Mechanisms / Brain Areas Involved

Brain area	Function
Prefrontal cortex	Working memory
Hippocampus	Formation/consolidation of new memory

Brain area	Function
Neocortex	Long-term storage
Striatum + motor cortex + cerebellum	Procedural memory
Amygdala	Emotional component of memory

5. Habituation

- Repeated application of a **neutral stimulus** → progressively **decreased response**.
- Mechanism:

Repeated stimulus

↓

↓ **Serotonin**

↓

↓ **Ca²⁺ entry into presynaptic terminal**

↓

↓ **Neurotransmitter release**

↓

Decreased response

6. Sensitization

- Repeated/noisome stimulus → **increased response**.
- It occurs by **presynaptic facilitation**.

Noxious stimulus

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↑ **Serotonin**

↓

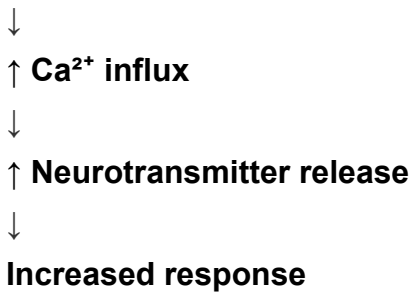
↑ **cAMP**

↓

K⁺ channel closure

↓

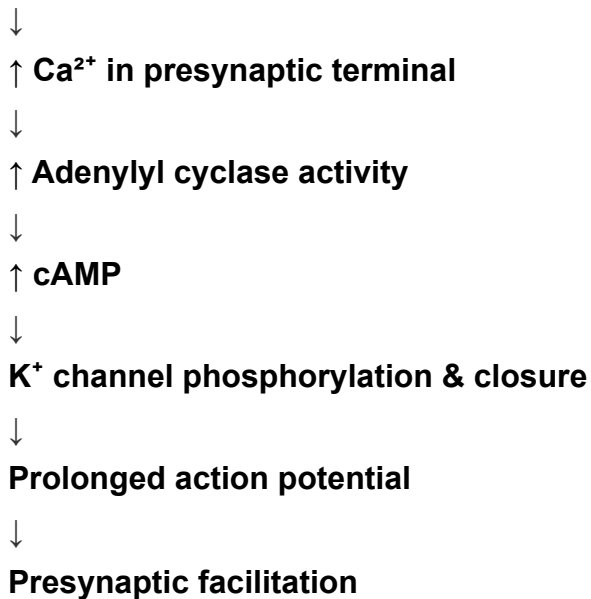
Prolonged action potential



7. Classical Conditioning

- Pairing of **UCS with CS** produces biochemical changes in target neurons.
- Basic mechanism involves **presynaptic facilitation**.

UCS + CS



8. Post-tetanic Potentiation

- **Post-tetanic potentiation** = enhanced postsynaptic potentials following a brief high-frequency (**tetanizing**) stimulation.
- Tetanizing stimulation → **Ca²⁺ accumulation in presynaptic neuron** → enhanced neurotransmitter release.
- Effect lasts for **up to about 60 seconds**.

9. Declarative and Procedural Memory

Declarative memory

- Memory of **facts and events**.
- Mainly involves **medial temporal lobe**.

Procedural memory

- Memory of **skills and habits**.
- Involves:
 - Striatum
 - Motor cortex
 - Cerebellum

10. Role of Cholinergic Neurons

- **Acetylcholine** is an important neurotransmitter in learning and memory.
- Cholinergic neurons in the **basal forebrain** project to:
 - Hippocampus
 - Cerebral cortex
- Loss of cortical/subcortical cholinergic neurons is associated with **memory impairment and dementia**.